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Identity Platforms and Anti-LGBTQ+ Legislation: Implications for Safeguarding Personal Data

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Abstract. Little is known about how digital identity platforms can be managed and operated in countries where governments have introduced anti-LGBTQ+ legislation. This work-in-progress study explores the tensions associated with digital identity platforms in these contexts. It draws on concepts from actor network theory to analyze field data collected by the authors in the Health Information Systems (HISP) research programme around a digital platform (DHIS2) that is installed in 114 countries globally, primarily in the health sector. The study identifies contradictory interests among the human and non-human actors, and analyzes these according to (1) the operation of a digital identity platform in a legally gray area, (2) visibility and invisibilization of vulnerable populations, and (3) human rights vs neo-colonialism. In particular, the paper identifies the potential for enrollment of the identity platform technology from intended into realities of use in identifying LGBTQ+ individuals and groups, identifying risks of data-induced harm and routes to overcoming them.

Keywords: Digital identity · Digital platforms · LGBTQ+ · vulnerable populations · Actor Network Theory

1 Introduction

Digital identity platforms are digital platforms with the ability to convert individuals into machine-readable data [1]. Such systems have a fiduciary responsibility to safeguard personal data for these individuals. Challenges emerge in contexts where the exposure of such data risks the prosecution of individuals based on their sexual orientation or gender identity. Sixty-two countries have anti-LGBTQ+ (lesbian, gay, bisexual, transgender, queer/questioning, other) legislation that criminalizes people who have been identified to engage in same-sex relations. Little is known about how managers of digital identity platforms should respond to this challenge. Academic research is limited in offering

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theory-informed knowledge and without this systematic knowledge, it is unclear how an identity platform can and should be managed in this context. This creates the risk of ad-hoc decision-making based on misconceptions about the concerns of members of the LGBTQ+ communities. To address this gap, our study is guided by the following research question:

RQ: How can digital identity platforms be managed where the exposure of the identity data could potentially be used to prosecute LGBTQ+ people?

To approach this question, this work-in-progress draws on actor-network theory (ANT) [2] to unpack the human and non-human stakeholders' interests in a case study and the potential for enrollment of the identity platform technology into alternative actor networks. Our empirical focus is on the Health Information Systems (HISP) research programme around a digital platform (DHIS2) installed as a national health management information system in 114 countries globally, including many of the countries where LGBTQ+ identities are criminalized.

The actor network analysis finds contradictory actor interests and analyzes them around three main areas: (1) operating in a legally gray area, (2) the processes that lead to visibility and invisibilization, and (3) human rights vs neo-colonialism. These issues are framed in opposition as 'digital colonialism', a practice where digital tools are plied to the systematic exploitation of structurally vulnerable populations [3, 4]. The paper presents work in progress towards future work on actor-network analysis of operating digital identity platforms. It seeks to understand the challenges and build systematic knowledge that can be used to develop operating policies and practices for platforms to prepare for and respond to such legislative changes in countries where they operate.

2 Background

2.1 Safeguarding Data

Managers of digital identity platforms have a fiduciary responsibility to safeguard personal data for all people represented within their databases [5]. The exposure of such data can have serious implications for the safety of individuals [6] and legislation mandates that managers maintain secure data practices to mitigate the risk of data breaches [7]. Little is known, however, about the management of extraordinary situations when governments demand data relating to people who are breaking the law. In the past decade, there has been an increase worldwide in government demands for access to data held by the private sector, a practice referred to as systematic government access [8]. The legal obligation to adhere to the law of the land, however, does not always synchronize with the moral obligation, and circumstances where laws are deemed to be unethical present challenges for ethical practices for digital identity platforms [9], as is the case where a country enacts anti-LGBTQ+ legislation.

2.2 Anti-LGBTQ+ Legislation Worldwide

Sixty-two countries have legislation in place that criminalize the lives of LGBTQ+ people [10], and there are many cases where LGBTQ+ people in these countries have

been convicted for engaging in same-sex relations. Gay couples have been arrested and imprisoned in many countries where the DHIS2 platform is utilized for national health surveillance [11–14]. Such anti-LGBTQ+ legislation has been widely condemned by the international community as a violation of human rights [15] which is 'contrary to … commitments on dignity and non-discrimination, and the prohibition of cruel, inhuman or degrading punishment' [16]. Such challenges point to hotly debated topics on the universality of human rights, particularly LGBTQ+ rights, with division ranging from the portrayal of such rights 'carrying hopes for freedom' and an aspiration for global connections, to the portrayal of them as 'a frightening new force of global coercion' [17]. Challenges around anti-LGBTQ+ laws have serious implications for organizations involved in the development and use of digital identity platforms for collecting, storing, and sharing personally identifiable data. While the security measures around data handling typically ensure that personal data does not fall into the wrong hands [7], in the countries that criminalize LGBTQ+ lives, the legislation dictates that they must be identified, and criminal proceedings be brought against them. The enforcement of such laws requires evidence to prove that an individual is engaging in these criminalized acts. Digital identity data and health data are potential sources of such evidence. Routine health data can potentially be used to infer whether a person is a member of the LGBTQ+ communities based on their behavior and practices. Such data can be derived from health records, from population census data, and from public health surveillance [18]. This creates a difficult management problem for digital platform designers and operators.

3 Theoretical Framework

Actor-network theory, or ANT [2] is a suitable frame for our inquiry because technology is viewed as much a product of social construction as of technical innovation. We draw on ANT as a sensitizing framework for storytelling about networks of actors - both human and non-human - and about the processes by which technology is established and entangled into the interests of actors and networks [19]. ANT enables focus on the stakeholders, or actors, within the socio-technical network and how they are shaping the forms of usage and social spread of the technology. The interests of the actors may vary widely and may encourage or constrain the technology. The process of alignment of actors' interests or shifts in alignment with the actor network occurs as actors enroll others into the network. ANT posits that aligning interests of actors in the network involves translation into common interests in using the technology achieved through inscriptions. Inscription may be used to support the translation of interests supporting the actor's interests and hence the aligning of the actor network. Table 1 below summarizes the main concepts of relevance to our inquiry.

Table 1. Summary of salient key concepts/ideas in actor network theory

Concept/idea	Definition/explanation
Actor	ANT focuses on stakeholders (actors) within a socio-technical network and how they are involved in shaping the form, adoption, innovation and geography of a technology. May include non-human stakeholders such as technology and texts [20, 21]
Actor network	Heterogeneous network of aligned interests, including people, organizations and standards [22]
Translation and inscription	The interests of the actors may vary widely and may encourage or constrain the technology. Actors' interests in the networks are translated into interests through common definitions, meanings and inscriptions attached to the technology [23, 24]
enrollment	The process of alignment of actors' interests with the actor network occurs as actors enroll others into the network [21]
Alignment	Alignment is concerned with the extent to which the agendas and interests inscribed into the practices, institutions, and strategies of the network pull in the same direction and serve the same purpose. Lack of alignment occurs when agendas and interests are diverging, in opposition, or competing. The political economy of competing agendas is concerned with the issue of whose network will dominate. [23, 25]

4 Methodology

This interpretive study [26, 27] draws on data from the authors' individual and collaborative activities concerning the safeguarding of personal data within implementations of DHIS2. The lead author conducted two months of ethnographic fieldwork and 15 interviews in India in 2023 with activists, health and welfare providers, and gender diverse people, analyzing how the transgender category is mobilized and understood in North India regarding health and welfare service provision. The third author's engaged scholarship [28] with DHIS2 implementation spans multiple country contexts over more than ten years. Participant observation has been undertaken during software development at HISP UiO, discussions with funders, and participation in or running training workshops, involving practical engagement in the discourses between various stakeholders regarding DHIS2 policies on privacy, security, confidentiality, and digital identity. The other two authors' have supervised DHIS2-related research projects for masters and PhD students.

For this study, the data was combined during intensive discussion and analyzed into themes derived from inductive analysis of the authors' respective fieldwork experiences and sensitized using actor-network theory concepts, producing the list of human and non-human actors. This enabled the study to make sense of the relevant influences, particularly networks, interests, translations, and inscriptions. Applying a case vignette approach allowed for the research to focus in depth on issues that were raised by the imposition of the strictest anti-LGBTQ+ laws across the experiences of the authors while keeping the specific country anonymous. This approach was taken because the

research topic raises serious concerns for the safety and security of individuals and thus is conducted with concern for privacy and ethical research practices.

5 Case Description

HISP is a network of North-South-South collaborations where the University of Oslo, Norway (UiO) has a key role in coordinating the network. The District Health Information System 2 (DHIS2) platform is a free and open-source platform used in 114 countries worldwide. It has a core database, a suite of generic applications that cover data entry, analysis, and system administration, and an application programming interface (API) [29]. Its core development is based at the HISP Centre at the University of Oslo (HISP UiO), Norway, and each individual implementation of the platform is independently owned and implemented by ministries of health.

In this study, *HISP Organization* is a local HISP team within the HISP network. It operates a DHIS2 implementation in an African country that is a former colony where the parliament has passed legislation that criminalizes same-sex relations. People known to be engaging in such relations face prosecution, including imprisonment or the death penalty.

The DHIS2 implementation is managed by the HISP Organization actor network, which includes the HISP employees in the Ministry, and the HISP Centre at UiO. HISP UiO acts as a conduit for financial and project support and capacity for the HISP Organization, responding to the needs with new features and technical solutions. In turn, HISP Organization champions the use of DHIS2, develops new implementations, and feeds back innovation and best practices to HISP UiO. To be officially recognized by HISP UiO, the HISP Organization has signed a memorandum of understanding (MoU) with UiO.

The DHIS2 technology is a non-human actant that contains the potential to be locally configured to capture data that could be used to identify LGBTQ+ individuals and their addresses. The implementation used by HISP Organization is inscribed with the design principles from UiO and operates in the Ministry of Health in the host country. This inscription is in the form of an open-source license. DHIS2 is also used by NGOs that target outreach and clinical services to key populations such as men who have sex with men and sex workers. These NGOs utilize DHIS2 to track their own program. The data remain independent and are not shared with the MoH or any other government organization.

The relevant actor networks are outlined in Fig. 1 below:

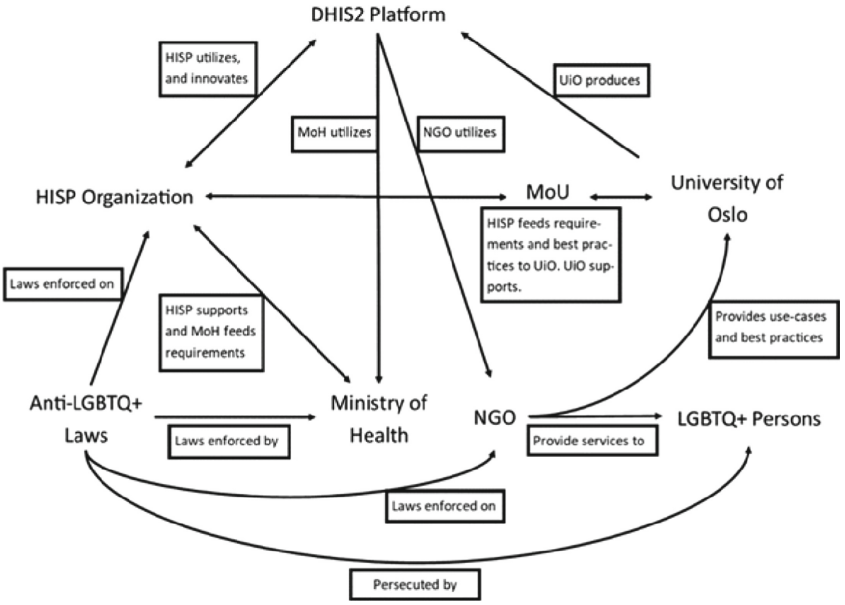


Fig. 1. Relevant actor networks

6 Analysis

6.1 Operating in a Legal Gray-Area

The HISP Organization actor network contains NGOs providing clinical services to LGBTQ+ and key populations. The DHIS2 Platform non-human actant negotiates the potential that the platform may be required to enforce the anti LGBTQ+ legislation. The actors in this network may decide collectively to safeguard LGBTQ+ identities while operating in a legal 'gray' area. Such a decision challenges the operating policies and governance of the UiO embedded in DHIS2 where the organization is operating against the law, even if the law is not being enforced.

6.2 Visibility vs Invisibilization

The collection of identity data risks the exposure of LGBTQ+ people to prosecution. However, the practices of hiding data that can identify LGBTQ+ people are translations that may lead to the long-term invisibilization of the LGBTQ+ communities, rendering them invisible to policy decisions. To carry out public health surveillance, data is needed to monitor disparities within the population over a long period of time. By halting the collection of this data, it becomes impossible to accurately recognize how determinants of health are affecting members of the LGBTQ+ communities. While HISP Organization or NGOs may choose to respond to the anti-LGBTQ+ law by halting the collection of all data that could make individual LGBTQ+ people identifiable, such a move would render the population invisible, amplifying existing inequities for the community in areas such as health and welfare.

6.3 Human Rights vs Neo-Colonialism

The HISP Organization actor network extends beyond the borders of the host nation to University of Oslo (UiO) Norway where the platform is produced. Tensions emerge between HISP Organization, the other human actors, and the perceived values inscribed within the non-human actants.

The host nation, whose government is within HISP Organization's actor network, is a former colony and, since gaining independence, has carried out efforts to build a national identity. As with many other nations who have enacted anti-LGBTQ+ legislation, the presence of LGBTQ+ identities is seen as an imposition of Western values, and the legislation in this study can be interpreted as part of a post-colonial project to reject the universality of Western values and a retaliation against a 'new force of global coercion' [17]. Concurrently, UiO, another actor in the HISP Organization actor network, upholds the Universal Declaration of Human Rights. However, the Memorandum of Understanding (MoU), a contract document between UiO and HISP Organization, is not explicitly inscribed with the promotion of the security and safety of LGBTQ+ people, and this giving space for interpretation by HISP Organization.

HISP Organization's response to this, through their practices and inscriptions in the DHIS2 implementations, influences how they are perceived by other actors in their actor network. On the one hand, a refusal to reveal the personal identity data of LGBTQ+ people on demand from law enforcement officers would be an alignment with the UiO actor network. On the other hand, it may be perceived by the host nation government's actor network as an imposition of Western values, and the inscription of values into the DHIS2 platform that contradicts their own. This could be perceived as an act of neo-colonialism in a country with a difficult history of occupation. Navigating this tension requires systematic knowledge and a clear pathway to address these tensions.

7 Conclusion and Anticipated Contribution

This research in progress study seeks to explore how digital identity platforms can be managed where the exposure of identity data could potentially be used to prosecute LGBTQ+ people. Thus far, using a case study and applying actor-network theory as an analytical tool, the study has exposed three areas where challenges arise.

We anticipate the study findings to have practical implications for digital identity platforms operating in countries where LGBTQ+ relationships are criminalized. We also anticipate a theoretical contribution to the fields of ICT4D, and to wider discourse in the data justice and surveillance studies fields by illuminating how data-induced harm plays out through the potential exposure of personally identifiable data.

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